Pulses in Cables - Lab Report

Leon Qian

Terry Chan

November 13, 2024

Abstract

In this lab, transmission lines are modelled as a series of chained LC units. Using this model, experiments measuring the transmission delay, signal losses, and behavior of pulses in transmission lines were performed. Using a waveform generator and oscilloscope, pulse reflections under different load conditions were measured and the data obtained was used to calculate signal propagation speeds. Results were compared to predictions using theoretical models to assess the accuracy of transmission line models and understand how cable properties affect signal transmission.

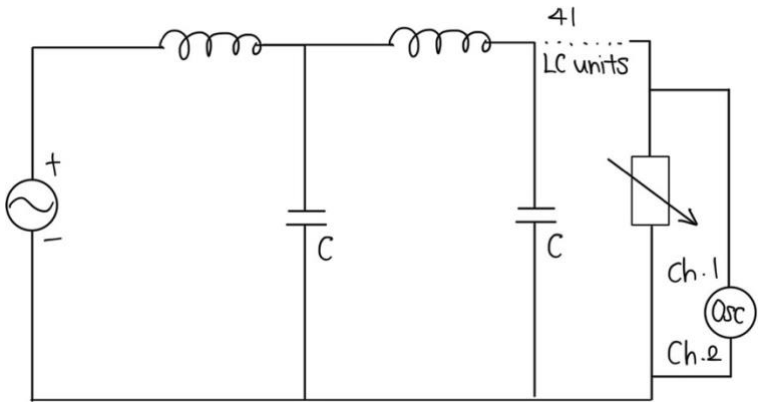
Introduction

Electrical signals in transmission lines like coaxial cables travel at finite speeds, unlike the instant response assumed in basic circuit theory. The speed and behavior of these signals are influenced by the cable's physical properties, such as inductance, capacitance, and insulation. This experiment investigates how pulses propagate through transmission lines by using LC circuit models. We focus on measuring signal speed, impedance matching, and reflections. By introducing impedance mismatches and observing the resulting pulse reflections with a waveform generator and oscilloscope, we aim to determine propagation speeds and compare them with theoretical values.

Theory

In this experiment, we analyze the propagation of electrical pulses along transmission lines, involving both coaxial cables and discrete LC circuit models. Transmission lines are modeled as repeated units of inductors (L) and capacitors (C).

Circuit diagram:



Equations:

The speed of wave propagation $v^2 = \frac{1}{L_0 C_0} \Rightarrow v = \frac{1}{\sqrt{L_0 C_0}} = \frac{1}{\sqrt{\varepsilon\mu}}$, where $L_0 = \frac{dL}{dx}$ and $C_0 = \frac{dC}{dx}$, ε and μ are the permittivity and permeability constants of the cable, respectively. (1)

The attenuation factor formula:

Attenuation(dB) = $10\log_{10}\frac{OutputIntensity(W)}{InputIntensity(W)} = 10\log_{10}\left(\frac{V_{relfected}}{V_{initial}}\right)^2$ (2)

Linear model fit: $y = ax + b$, where a, b are constants to be determined from the fit. (3)

Procedures

Part 1

Begin by connecting the waveform generator to Channel 1 of the oscilloscope and, via a T-connector, to one end of the transmission line consisting of 41 LC units (refer to the circuit diagram in the theory section). Attach the output ends of the transmission line to a variable resistor. Power on the oscilloscope and the waveform generator. Configure the waveform generator to emit square pulses with a precise width of 75 μs and a frequency of 300 Hz. Introduce a resistive load R across the transmission line and adjust it until the return pulse is no longer observable (the pulse should appear as a flat line), indicating an impedance match. Record this resistance value. To stabilize the oscilloscope display, set the trigger mode to Channel 1 on a rising edge with a threshold of +2.00 V. Adjust the horizontal acquire mode to "Averaging" with

the number of averages set to 65536 (which is the maximum number on the oscilloscope). Set the time reference to "Left" to align the waveform at the start of the display. Configure the vertical scale for Channels 1 and 2 to 1.40 V/division and the horizontal scale to 30.0 μs/division, then zero both axes. For measurement settings, configure the oscilloscope to measure "Delay" (response time) using Channel 1 as the source. Utilize the "Measuring Edge" setting with thresholds at the lower (10%), middle (50%), and upper (90%) levels to accurately capture pulse timings. Ensure all measurements are taken under these settings to maintain consistency and accuracy throughout the experiment.

Part 2

To conduct the second part of the experiment, configure the waveform generator to emit square pulses with a frequency of 15 kHz and a pulse width of 50 ns, keeping the amplitude at the default 4.000 V peak-to-peak (Vpp). Connect the waveform generator according to the circuit diagram provided in the Theory section. Adjust the oscilloscope settings by setting the detection thresholds to lower (2%), middle (5%), and upper (10%), and the trigger threshold to -1.89 V. To obtain a suitable display window for measurements, set the horizontal offset to 12.00 ns, the horizontal scale to 12.00 ns/division, and the vertical offset to 2.00 V. Use the oscilloscope's magnification function to zoom in for a clear view of the waveform. Add the "Period" parameter to the measurement screen to monitor the pulse delay time accurately. Reset the statistics to synchronize the amplitude and period counts and disable the relative sigma option to display the absolute standard deviation. In the cursor menu, select the cursor mode "Track" to follow the waveform and set the X_1 and X_2 cursors to "Linked," ensuring their difference matches the previously measured period; the ΔY represents the voltage difference between the cursors. These ΔX and ΔY values are used to calculate the attenuation factor and signal delays.

Uncertainties

Uncertainties in the measurements made using the oscilloscope were determined based on specifications provided in the datasheet, taking into account the horizontal and vertical scales as well as the offsets. Additionally, the given uncertainties for the capacitance and inductance of each LC unit were incorporated into the calculations. Beyond these factors, no other significant sources of error were identified that would have a meaningful impact on the experimental results.

Results

Part 1

Impedance match: 385 Ω (determined by the process in the methods section)

Capacitance of each LC-unit = 0.01μF ± 3%; Inductance = 1.5 mH ± 10%. Calculated theoretical speed of wave propagation by equation (1): $(2.582 \pm 0.135) \times 10^5$ LC units/s.

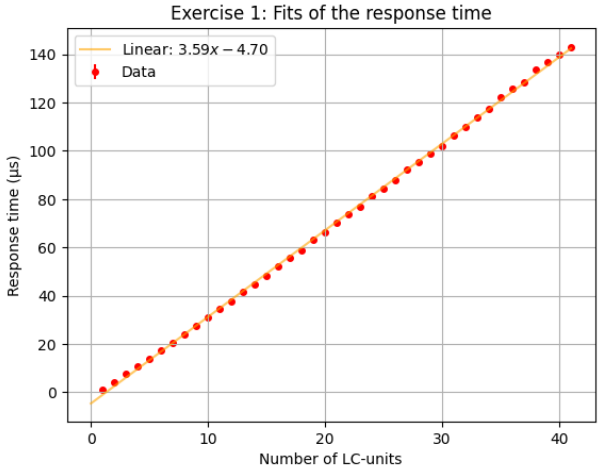


Figure 1: Plot of the response time versus the number of LC-units with data points and a linear model fit.

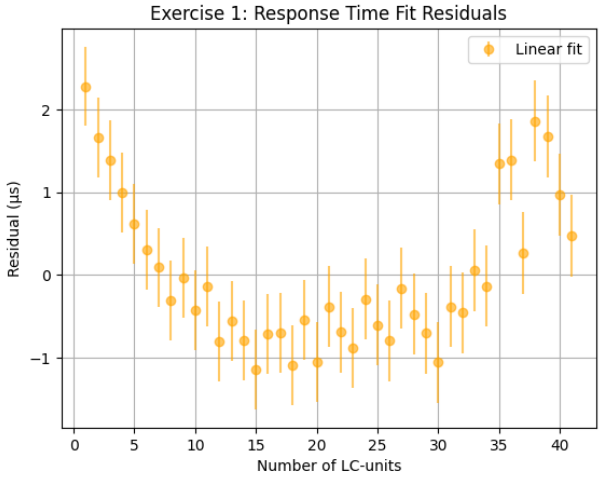


Figure 2: Plot of the residuals between the observed data and the linear fit, with error bars indicating uncertainties.

Based on equation (3), the fitted linear model is given by: $y = (3.590 \pm 0.006)x - (4.699 \pm 0.154)$. The chi-squared value for this fit is 34.15 ± 8.83, with a reduced chi-squared of 0.88 ± 0.23, indicating a fair agreement between the model and the experimental data. The fitted speed of wave propagation is $(2.785 \pm 0.005) \times 10^5$ LC units/s. In comparison, the experimental speed is slightly higher than the theoretical prediction, but the values are reasonably close, given the uncertainties.

The residuals in the plot show potential higher-order behavior not captured by the model. This may be the result of underlying effects not accounted for by the simple linear model or variations in experimental conditions introducing systematic biases. Additionally, imperfections in the experimental setup, such as slight deviations in the characteristics of the LC units or external influences could contribute to these deviations. The reduced chi-squared value of 0.88 is close to 1, suggesting that the linear model provides a reasonable fit.

Part 2

Theoretical transmission speed of the cable by equation (1): 1.999×10^8 m/s

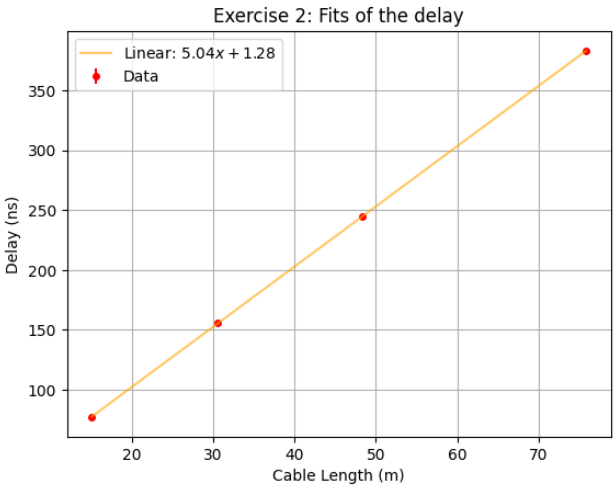


Figure 3: Plot of the response time versus the length of cable with data points and a linear model fit.

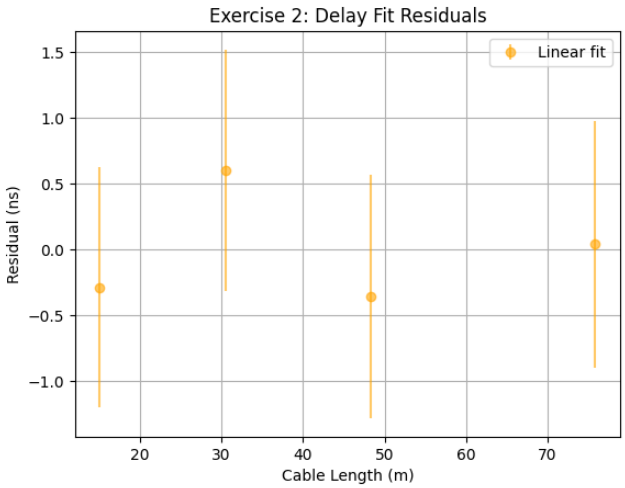


Figure 4: Plot of the residuals between the observed data and the linear fit, with error bars indicating uncertainties.

From equation (3), the fitted linear model is given by $y = (5.037 \pm 0.020)x + (1.283 \pm 0.974)$. The chi-squared value for this fit is 0.57, with a reduced chi-squared of 0.28, indicating a strong agreement between the model and the experimental data. This is confirmed by the residual plot, where the residuals are randomly distributed around zero, with no significant patterns indicating systematic errors. The transmission speed of the cable calculated from the fit is $(1.985 \pm 0.008) \times 10^8$ m/s.

The strong linear relationship between delay and cable length is consistent with theoretical predictions, which suggest that delay should increase linearly with the length of the cable.

From the data, and using eq. (2), the attenuation factor of the cable is calculated as $(4.177 \pm 1.276) \times 10^{-2}$ dB/m.

Conclusion:

This lab investigated the propagation of electrical pulses in transmission lines using LC circuit models and coaxial cables. By analyzing the relationship between delay and cable length, we observed a strong correlation, consistent with theoretical expectations. The calculated propagation speed aligned reasonably well with theoretical values, though some deviations were noted, likely due to experimental imperfections and equipment limitations. Impedance matching was crucial to minimize reflections, and uncertainties were carefully considered to validate our findings. Overall, the linear model effectively described pulse propagation, and the results provided a solid understanding of how signal transmission behaves in practical setups.

From the analysis, we obtained a signal transmission speed in the repeated LC-unit model as $(2.785 \pm 0.005) \times 10^5$ LC units/s in Part 1. The transmission speed of the transmission lines was measured to be $(1.985 \pm 0.008) \times 10^8$ m/s, and the attenuation factor of the cable was measured to be (4.177±1.276) $\times 10^{-2}$ dB/m in Part 2. The characteristic impedance of the cable was given to be 50 Ω. Overall, the results appear to be reasonable and match theoretical values.